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Online Federated Learning on Resource-Limited IoT Devices for Traffic Flow Prediction in Smart Mobility Ecosystems

Marco Pizzolante[1], Angela-Tafadzwa Shumba[1,2], Teodoro Montanaro[1], Gianluigi Semeraro[1], Davide Cantoro[1]
Davide Rollo[1], Mattia Cotardo[1], Ilaria Sergi[1], Paolo Visconti[1], Luigi Patrono[1,*]

[1]*Dept. of Engineering for Innovation, University of Salento, Lecce, Italy*
[2]*Istituto Italiano di Tecnologia (IIT), Lecce, Italy*

{marco.pizzolante, angela.shumba, teodoro.montanaro, gianluigi.semeraro, davide.cantoro
davide.rollo, mattia.cotardo, ilaria.sergi, paolo.visconti, luigi.patrono}@unisalento.it

***Abstract*—Traffic congestion significantly impacts safety and urban livability in smart cities, motivating the development of accurate Traffic Flow Prediction (TFP) systems. Traditional deep learning approaches typically rely on centralized training, which is difficult to scale in distributed Internet of Things (IoT) environments. To address these limitations, decentralized paradigms such as Local Learning (LL) and Federated Learning (FL) enable on-device training and collaborative model updates while preserving data locality. For real-time TFP, the inherently non-stationary nature of traffic data necessitates continuous model adaptation, making online federated learning essential for scalable and collaborative deployment. However, this setting remains challenging because traffic data are typically non-IID across clients, with local patterns varying significantly across locations and devices. This paper presents an exploratory study of online federated learning for TFP on resource-constrained IoT devices. Using a GRU-based network as a common backbone, the Online Federated Learning paradigm is benchmarked relative to Centralized and Local Learning as reference baselines. A performance evaluation was conducted by evaluating RMSE and MAE on the PEMS-BAY dataset. Robustness to non-IID data is further assessed using FedProx and SCAFFOLD. Results show that LL achieves the lowest prediction error, whereas FL degrades as the number of local epochs increases, and non-IID mitigation strategies provide limited improvements under low-latency constraints. Overall, online federated learning is a viable approach for real-time TFP, but its performance is highly sensitive to client heterogeneity.**

***Index Terms*—Traffic Flow Prediction, Decentralized Learning, Federated Learning, Online Learning, Internet of Things.**

I. Introduction

By 2030, nearly 60% of the world population is projected to live in urban areas, highlighting urbanization as one of the most rapidly advancing global phenomena [1]. This poses significant challenges to both environmental sustainability and residents' quality of life. In this context, the development of Traffic Flow Prediction (TFP) systems emerges as a key component of future smart cities to alleviate road congestion, increased accident rates, additional fuel consumption, and excessive air pollution [2], [3]. At their core, TFP systems model road networks by collecting historical traffic data and applying Deep Learning (DL) techniques to predict future traffic conditions, thereby enabling proactive mobility management and routing decisions. In line with this, the Internet of Things (IoT) enables distributed sensors, referred to as traffic nodes (or equivalently edge nodes) in the TFP literature, to acquire real-time roadway data that supports on-device training and inference at the edge [4]–[6]. Traditionally, TFP models have been trained in a centralized manner, where large volumes of raw data must be transmitted to a central server, incurring substantial communication overhead and storage costs [7]. To overcome these limitations, decentralized learning has attracted extensive research interest in both academia and industry, as it allows edge nodes to retain raw data locally and train models on-device, bringing learning closer to the data source. Among decentralized approaches, two primary alternatives have emerged: Local Learning (LL) and Federated Learning (FL). As shown in Fig. 1, which graphically distinguishes the three alternative characteristics, LL represents the most naive alternative, adopting a fully decoupled paradigm where each node trains its model in isolation, with no parameter exchange across the network [8]. Federated Learning, on the other hand, introduces a coordinated aggregation mechanism in which client-side model updates are periodically transmitted to a central server, aggregated into a global model, and redistributed to all participants. Despite its promising advantages, FL is still a relatively new training paradigm. As it continues to evolve, further research is needed to thoroughly evaluate its capabilities and limitations and to address practical challenges such as statistical heterogeneity.

In smart-mobility environments, traffic nodes typically operate under different local conditions, resulting in non-Independent and Identically Distributed (non-IID) data [9]. In collaborative learning, this heterogeneity creates conflicts during the server-side aggregation phase, degrading model performance. Additionally, because traffic data evolves over time due to factors such as travel habits and the environment, TFP systems need to learn quickly and adapt continuously. Therefore, continuous-learning paradigms are best suited for this TFP scenario. Static models, on the other hand, are inherently vulnerable to concept drift, a phenomenon in which

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the progressive divergence between training and deployment distributions gradually undermines prediction reliability [10], [11]. Recently, the research community has been increasingly moving toward Online Federated Learning, yet its application to the TFP domain remains largely underexplored. This work presents an exploratory study investigating this gap. The study benchmarks Online Federated Learning for TFP relative to Centralized and Local Learning as reference baselines. The implemented models are deployed on real-world resource-constrained IoT devices and benchmarked for predictive accuracy, measured via Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). Additionally, the study explores the performance of FL under conditions of statistical heterogeneity. Two widely adopted non-IID mitigation strategies, namely FedProx [12] and SCAFFOLD [13], were implemented in an online learning framework. All experiments are conducted on the PEMS-BAY benchmark dataset [14] using the open-source FL framework Flower [15]. Collectively, the findings provide valuable insights into the practical viability of FL-based TFP systems for real-world smart-mobility deployments.

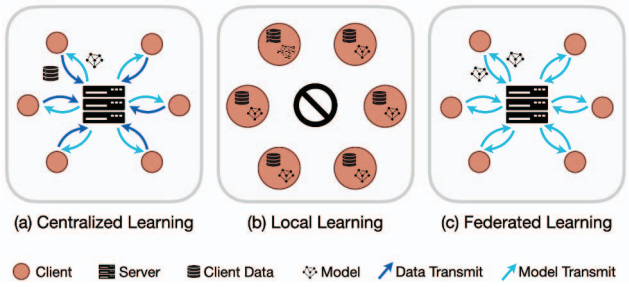


Fig. 1: Overview of centralized, local, and federated learning. (a) Raw data is collected at a central server–, which trains and distributes a shared global model. (b) Each client trains independently on local data, without any communication or data sharing. (c) Clients train locally and upload only model parameters to the server, which aggregates them and returns the resulting global model (therefore, no raw data is sent to the server).

II. Related Works

Research on TFP has undergone substantial evolution, progressing from classical statistical approaches to decentralized and drift-aware learning systems. Early forecasting work relied on classical parametric time-series models, most notably AutoRegressive Integrated Moving Average (ARIMA) and its variants [16], [17]. However, these typically assume stationarity and stable linear dynamics, which limits their effectiveness when forecasting highly fluctuating traffic flows. Consequently, machine learning and deep learning techniques became popular. Thanks to their ability to capture temporal patterns in traffic data sequences, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) have been widely adopted [18], [19]. Furthermore, numerous works utilize Graph Neural Networks (GNNs) and Graph Attention Networks (GANs) to exploit spatial correlations among nodes in the transport network [20]–[25]. A key limitation of these approaches is their centralized data aggregation, which imposes substantial communication overhead and scalability bottlenecks, making the resulting models unsuitable for real-time deployment. To overcome these problems, researchers began exploring Local Learning. For example, Qifeng Lai et al. and Yupin Huang et al. [26], [27] developed localized models to process data closer to the source, reducing the burden on central servers. However, fully decoupled local models cannot leverage knowledge from other devices, limiting their generalization capabilities. To combine the benefits of edge computing with the power of collaborative learning, recent studies have increasingly adopted the Federated Learning paradigm. Several works focus on protecting sensitive information during distributed training by leveraging spatiotemporal models, including GCN- and GRU-based architectures, in combination with secure aggregation mechanisms [28], [29]. Other studies address the protection of transportation network topology by applying differential privacy to local adjacency matrices [30]. Other researchers aim to alleviate communication overhead and improve scalability. In this context, dynamic client selection strategies have been proposed to optimize model updates and reduce unnecessary transmissions [31]–[34]. Additionally, the non-IID nature of client data has been extensively investigated in the FL literature, as it constitutes a primary source of convergence instability. The literature treats statistical heterogeneity through a variety of strategies, including client clustering [29] and knowledge distillation [35]–[37]. Beyond batch FL, researchers have explored Online Federated Learning to handle streaming data in scenarios where traditional offline FL models struggle with concept drift. Works addressing concept drift adaptation include FedOSTC and REFOL [10], [11]. FedOSTC combines GRU-based local encoders with a server-side Graph Attention Network, optimized via Online Gradient Descent, while REFOL builds on it by introducing a client participation mechanism based on KL Divergence to trigger local updates only when significant distribution shifts are detected, thereby reducing computational overhead. Another study adopts an online FL setting but shifts its focus toward the reliability of the aggregation infrastructure, leveraging a permissioned blockchain to orchestrate decentralized training over streaming data and eliminate any central point of failure [38]. Taken together, these studies establish a clear shift from batch federated learning to streaming, adaptive pipelines for TFP. However, three critical gaps emerge:

- existing work implicitly assumes that Federated Learning is the most suitable paradigm, revealing a lack of systematic comparisons across centralized, local, and federated architectures
- although numerous non-IID mitigation strategies have been proposed in the FL literature, their effectiveness in a continuous-learning setting remains largely unexplored
- most existing approaches rely on GPU-based client sim-

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ulations rather than real-world deployments, highlighting a limited evaluation of model performance on real IoT devices under practical constraints

As a result of these observations, this work empirically assesses whether Online Federated Learning is a viable architectural choice for real-world TFP systems, benchmarking its performance against alternative paradigms on physical IoT devices in heterogeneous non-IID scenarios to address these gaps identified in the literature.

III. Methods

In this study, roadside traffic speed data were handled as a stream, and online learning was used to update the model as data distributions changed over time, thereby addressing concept drift. In the online learning setting, each local update was set to $E = 5$ epochs to meet latency requirements and the computational constraints of IoT devices, in line with prior work [11]. A single-layer Gated Recurrent Unit (GRU) with 128 hidden units was used as the predictive model, as it provides competitive accuracy at lower computational, memory, and energy cost than more complex architectures. The model was trained and evaluated on PEMS-BAY, a widely used benchmark dataset comprising real-world traffic speed measurements recorded every five minutes by distributed sensors across the Bay Area over six months. To ensure a fair comparison across learning paradigms, all models were initialized using a shared offline pre-training phase on the first 90% of the dataset partitioned in a 7:1:2 ratio. This was done to reduce cold-start effects and to allow the online evaluation to reflect steady-state behavior rather than initialization artifacts. The remaining 10% of the dataset, corresponding to the most recent observations, was reserved for the online phase. Three learning paradigms were considered. In the federated setting, the pre-trained model was distributed to clients and refined through server-orchestrated communication rounds, in which locally updated models were aggregated to produce the next global model. The FedAvg [39] aggregation strategy was used as the baseline FL method. In the local setting, the same pre-trained model was deployed to each client and updated independently, without communication or parameter exchange. In the centralized setting, the model was trained offline on aggregated data and then used only for inference during the online phase, with no further updates. This setting served as the centralized baseline for the decentralized paradigms evaluated in this study. For the federated and local settings, the online phase was implemented using a sliding-window mechanism. At each time step, a one-hour window of past measurements was used to predict traffic conditions over the next thirty minutes. After the corresponding ground-truth values became available, the prediction error was computed and used to update the model incrementally. As illustrated in Fig. 2, the window is then advanced to incorporate newly arrived data, thereby emulating a continuous sensor data stream.

Predictive performance was evaluated using RMSE and MAE over a fixed online window of $\mathcal{T} = 576$, in line with prior TFP studies [10]. The experimental evaluation comprised

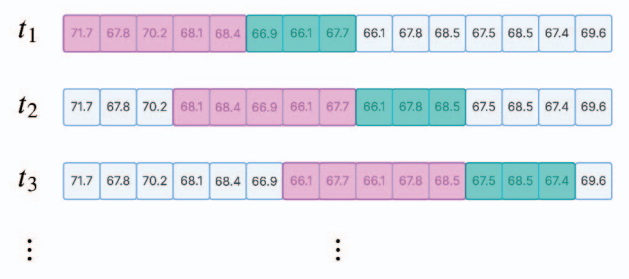


Fig. 2: Sliding window mechanism. The samples highlighted in magenta represent the input window used by the model to predict the subsequent future values highlighted in green.

two stages. The first evaluation benchmarked federated learning against local and centralized baselines as a function of the local epoch count E, with each online step consisting of a fast and lightweight local update of the pre-trained model. The second experimental phase was conducted to examine FL under highly heterogeneous conditions. In this phase, FedProx [12] and SCAFFOLD [13] were evaluated as alternatives to the baseline FedAvg strategy. FedProx constrains local updates through a proximal term, whereas SCAFFOLD reduces update bias through control variates. The FedProx proximal coefficient μ was selected through a preliminary grid search for each value of E.

IV. Implementation

The experimental framework was implemented in Python using PyTorch and Flower [15] and deployed via Docker [40] in a Kubernetes [41] cluster, with GitHub Actions [42] enabling automation and reproducibility. The deployment follows a client–server architecture, as illustrated in Fig. 3, with a central server coordinating training across distributed Raspberry Pi clients. The testbed comprised eight Raspberry Pi 5 devices with 8 GB of RAM, acting as edge clients, and a Dell PowerEdge T350 server with a 12-core processor and 64 GB of RAM, acting as the central coordinator. To align the data partitioning with this physical configuration, eight sensor columns were selected from the 325 available in the PEMS-BAY dataset and assigned to the eight Raspberry Pi clients. Each client was therefore associated with a unique local time series, reproducing the heterogeneous conditions observed across traffic nodes in real-world deployments. The selection procedure followed two criteria. First, the dataset was checked to confirm the absence of outliers. Sensors were then ranked according to the peak-to-peak amplitude of their traffic-flow values, and the highest-ranking candidates were retained to maximize variability across clients. Second, the GPS coordinates provided in the sensor metadata were used to ensure broad spatial coverage across the road network, so that the selected subset remained representative of the full dataset. Within the cluster, the server functioned as the control plane and the Raspberry Pi devices as worker nodes, enabling the orchestration of containerized machine learning workloads

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under realistic edge constraints. Flower was used to manage synchronous communication rounds and to distribute runtime configuration parameters to clients during execution. This deployment supported all three learning paradigms considered in the evaluation, while preserving a common infrastructure across experiments. Specifically, the centralized paradigm was implemented by deploying the pre-trained model to client devices for inference only, forecasting traffic speeds over $\mathcal{T}$ time steps without any further parameter updates. The implementation of the local paradigm involved deploying the same pre-trained model to each client, which independently updates its own copy as new measurements arrive. In this setting, the parameter exchange and aggregation strategies on the server are deactivated. Finally, the federated paradigm was instantiated using the Flower framework's built-in methods. To complement the deployment evaluation, client-side CPU and memory usage were monitored during the online federated learning experiments. Two auxiliary routines, implemented using the `psutil` library, were used to record CPU utilization and physical memory usage on each Raspberry Pi client. Measurements were acquired as instantaneous snapshots immediately after the optimization step in the model-update routine. The monitoring was performed over the 288 online federated rounds using FedAvg with $E = 5$ local epochs.

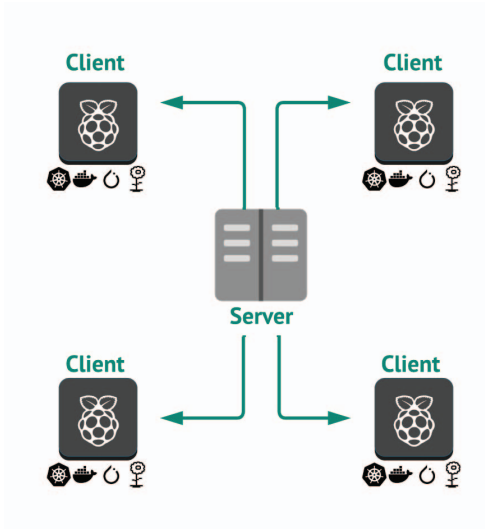


Fig. 3: System Architecture.

V. Results

The initial offline pre-training phase yielded an RMSE of 3.94 and an MAE of 1.91 on the held-out test set. The resulting weights were used to initialize all subsequent online experiments. The centralized static baseline achieved an RMSE of 2.57 and an MAE of 2.23. Local learning showed consistent improvement as the number of epochs increased. As shown in Fig. 4 at $E = 1$, the local models achieved an RMSE of 2.19 and an MAE of 1.87. Increasing the number of epochs to $E = 5$ reduced these values to 1.75 and 1.47, respectively, corresponding to an average relative improvement exceeding 30% across both metrics. In contrast, the federated baseline based on FedAvg worsened as E increased. At $E = 1$, FedAvg achieved an RMSE of 2.47 and an MAE of 2.15, corresponding to a modest relative improvement of 3.7% over the centralized baseline. However, at $E = 5$, performance degraded to an RMSE of 3.21 and an MAE of 2.88. The non-IID mitigation strategies yielded only limited improvements over FedAvg at low epoch counts, with no substantial gains for ($E \leq 2$). Their benefits emerged only at higher values of E. FedProx achieved an improvement of approximately 1.8% over FedAvg at $E = 5$ with $\mu = 0.65$. SCAFFOLD, on the other hand, produced larger gains, reducing the aggregated error by about 5.3% at $E = 4$ and 3.2% at $E = 3$ (Fig. 5).

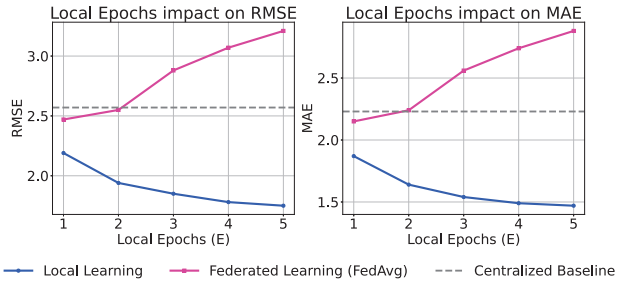


Fig. 4: RMSE and MAE as a function of local epochs for centralized, local, and FedAvg learning. The centralized model is shown as a fixed reference baseline (dotted line).

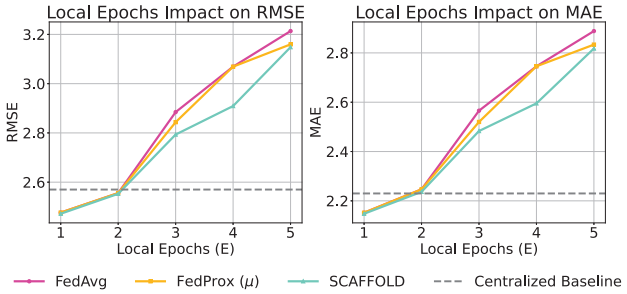


Fig. 5: RMSE and MAE as a function of local epochs for FedAvg, FedProx, and SCAFFOLD. The centralized model is shown as a fixed reference baseline (dotted line).

VI. Discussion

The results highlight a trade-off between predictive accuracy and learning paradigm. Although Local Learning achieved the lowest prediction errors, its practical deployment in real-world intelligent transportation systems is limited by its purely isolated nature. Local models are updated independently; therefore, they may become hyper-specialized to the local sensor dynamics and be unable to provide a shared network-wide representation for large-scale monitoring. Federated Learning addresses this limitation; however, the performance degradation observed with the baseline FedAvg algorithm at

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higher local-epoch counts highlights the impact of cross-client statistical heterogeneity (non-IID data). As E increases, local models drift toward different optima, making server-side aggregation less effective and ultimately degrading global model performance. This behavior is consistent with the gradient conflict problem reported in heterogeneous FL settings [43]. FedProx and SCAFFOLD provided only limited recovery, mainly at higher values of E, and remained ineffective under the low-latency and low-computation constraints of real-time streaming scenarios. These results suggest that algorithm mitigation alone may be insufficient. Additional architectural interventions to optimize network resource management and reduce heterogeneity may be required. Beyond the achieved performance results, the selected hardware proved capable of sustaining the full training workload without compromising system responsiveness. As shown in Figures 6 and 7, CPU and memory utilization remained stable across the online federated rounds, with no visible evidence of progressive CPU saturation or systematic memory growth. This evidence points to promising prospects for the deployment of online federated learning pipelines in real-world smart mobility ecosystems.

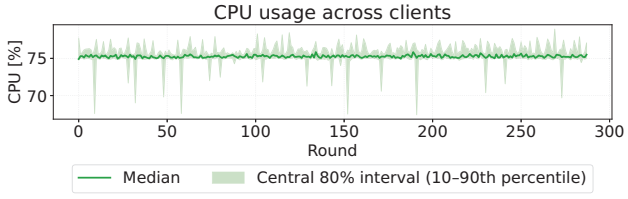


Fig. 6: Percentage of CPU usage across clients.

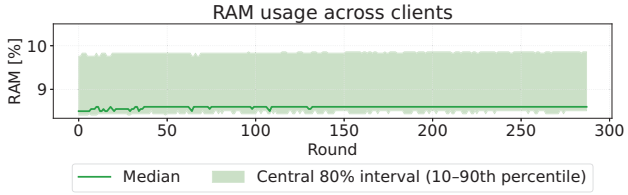


Fig. 7: Percentage of RAM usage across clients.

VII. Conclusions

This study investigated the practical capabilities and limitations of Online Federated Learning for traffic flow prediction in smart mobility ecosystems. Using a GRU-based prediction model, deployed on a set of real-world resource-constrained edge devices under realistic operational conditions. The results show a clear trade-off between predictive accuracy and system-level utility. Local learning achieved the lowest per-node prediction error but lacks the structural capacity to construct a network-wide representative model due to its isolated operation. FedAvg offers a more balanced alternative, but it remained highly sensitive to gradient conflicts induced by cross-client statistical heterogeneity, with performance degrading as the number of local epochs increased. FedProx and SCAFFOLD provided only limited improvements, particularly under the low-epoch regime required by online learning over continuous data streams. In light of these findings, future research should explore structural alternatives such as client clustering, which enables the training of specialized models on groups of nodes with similar traffic distributions, thereby reducing within-cluster variance without inflating the computational burden on edge devices. Asynchronous federated learning mechanisms and deep-sleep strategies can also be investigated to optimize dynamic client participation and improve the energy efficiency of federated IoT deployments.

Acknowledgment

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